THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 7th term is: 1

(a) 20 **(b)** 23 **(c)** 21 **(d)** 22

2. A complete graph has 3 points, each joined to every other point. How many lines are there? 1

(a) 3 **(b)** 6 **(c)** 2 **(d)** 4

3. The sum of the triangular numbers 10 and 15 is: 1

(a) 29 **(b)** 25 **(c)** 24 **(d)** 26

4. What is the sum of the first 15 natural numbers $1 + 2 + 3 + \cdots + 15$? 1

(a) 135 **(b)** 121 **(c)** 105 **(d)** 120

5. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 16th one is: 1

(a) 257 **(b)** 272 **(c)** 240 **(d)** 256

6. In the pattern $1, 3, 6, 10, \ldots$ what is the 9th term? 1

(a) 46 **(b)** 54 **(c)** 44 **(d)** 45

7. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 125 is: 1

(a) 216 **(b)** 226 **(c)** 215 **(d)** 217

8. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 8th term is: 1

(a) 34 **(b)** 35 **(c)** 36 **(d)** 33

9. A complete graph has 4 points, each joined to every other point. How many lines are there? 1

(a) 5 **(b)** 6 **(c)** 10 **(d)** 7

10. The sum of the triangular numbers 15 and 21 is: 1

(a) 37 **(b)** 36 **(c)** 35 **(d)** 41

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the first 10 terms of the Virahanka sequence that starts with 1, 1. 2

12. How many lines are needed to join all pairs among 8 points (a complete graph)? 2

13. Find the sum $1 + 3 + 5 + \ldots$ of the first 13 odd numbers without adding one by one. 2

14. Write the first 10 terms of the Virahanka sequence that starts with 1, 3. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Add the triangular numbers 45 and 55. What do you notice about the answer? 3

16. Write the sum $1^3 + 2^3 + \cdots + 3^3$ and show it equals the square of $1 + 2 + \cdots + 3$. 3

17. Look at the sequence $5, 7, 9, 11, 13$. Find the pattern and the 10th term. 3

OR

Add the triangular numbers 55 and 66. What do you notice about the answer? 3

18. Write the sum $1^3 + 2^3 + \cdots + 4^3$ and show it equals the square of $1 + 2 + \cdots + 4$. 3

OR

Look at the sequence $8, 11, 14, 17, 20$. Find the pattern and the 10th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A group of 6 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 5th and 6th terms are 8 and 13. Find the next three terms and add the first 5 terms. 5

20. A group of 7 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 6th and 7th terms are 13 and 21. Find the next three terms and add the first 6 terms. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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